

MTH203: Sequences and series

Tutorial/Homework 04

7/10/2025

Due date: 14/10/2025, 6:00 PM

Problem 1. Does the series $\sum_{n=1}^{\infty} \frac{1}{3n+1000}$ converge?

Problem 2. Prove that the series $\sum_{n=1}^{\infty} \frac{1}{n^n}$ converges.

Problem 3. Let $(a_n)_{n=1}^{\infty}$ be a sequence such that $a_n \geq 0$ for all n . Let be $(a_{n_k})_{k=1}^{\infty}$ a subsequence of this sequence. Prove that if $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{k=1}^{\infty} a_{n_k}$ also converges.

Problem 4. Prove that the series $\sum_{n=1}^{\infty} \frac{1}{n^2-20}$ converges. (Hint: First prove that $n^2 - 20 > (n-1)^2$ if n is large enough.)

Problem 5. Let the sequence $(x_n)_{n=1}^{\infty}$ be defined by $x_1 = \sqrt{2}$ and $x_{n+1} = \sqrt{2 + x_n}$ for $n \geq 1$. Thus, the first few terms are as follows:

$$\sqrt{2}, \sqrt{2 + \sqrt{2}}, \sqrt{2 + \sqrt{2 + \sqrt{2}}}, \dots$$

Does this sequence converge? (Hint: First observe that the sequence is strictly increasing. Now, show that it is also bounded by proving that $x_n \leq 2$ for all n . You can prove this by induction on n . Use this to conclude that the limit of this sequence exists. Prove that if the limit is L , then $L^2 = 2 + L$. Now calculate L .)